

SECI1143

PROBABILITY & STATISTICAL DATA ANALYSIS

Random Variables & Probability Distribution

Random Variables

- A numerical variable whose value depends on the outcome of a **chance experiment**.
- It associates a numerical value with **each outcome** of a chance experiment.
- Two types of random variables
 - 1) Discrete
 - 2) Continuous

Discrete Random Variables

- Discrete – its set of possible values is a collection of isolated points along a number line.



This is typically a “count”
of something

- Continuous - its set of possible values includes an entire interval on a number line.



This is typically a
“measure” of something

Example

Suppose that a counsellor plans to select a **random sample** of 50 seniors at a large high school and to ask each student in the sample whether he or she plans to attend college after graduation. The process of sampling is a **chance experiment**. The **sample space** for this experiment consists of all different possible random samples of size 50 that might result (there is a very large number of these) and for simple random sampling, each of these outcomes is **equally likely**.

Let x is the random variable.

- x represent the number of successes in the sample (who plans to attend college after graduation).
- Thus, x is discrete random variable (counting).



Example

A point is randomly selected on the surface of a lake that has a maximum depth of 100 feet. Let y be the depth of the lake at the random chosen point.

- The possible value of y : measurement in feet of the depth of the lake.
- Thus, y is continuous random variable.



Example

Consider an experiment in which the type of book, print (P) and digital (D), chosen by each of three successive customers making a purchase from on an online bookstore is noted. Define a **random variable** x by

x = number of customers purchasing a book in digital format

The experimental outcome can be abbreviated, e.g., DPD (first and third customer purchase a digital book).

Outcome	PPP	DPP	PDP	PPD	DDP	DPD	PDD	DDD
x value	0	1	1	1	2	2	2	3

There are only 4 possible x values: 0, 1, 2 and 3. Thus, x is discrete random variable.

Probability Distribution For **DISCRETE** Random Variables

Definition

- Probability distribution is a model that describes the long-run behaviour of a variable.
- The mathematical definition of a discrete probability distribution is : $p(x)$
- If $x = 2$, we write $p(2)$ in place of $p(x=2)$
- $p(5)$ denotes the probability that $x = 5$.

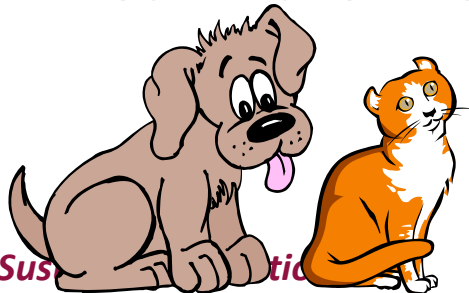
- It give the possibility associated with each possible x value.
- Each probability is the long-run relative frequency of occurrence of the corresponding x value when the chance experiment is performed a very large number of times.
- Common way to display a probability distribution for **discrete** random variable: **table**, **histogram**, **formula**.

Example

In a Wolf City, regulations prohibit no more than five dogs or cats per household.

Let, x = the number of dogs and cats in a randomly selected household in Wolf City.

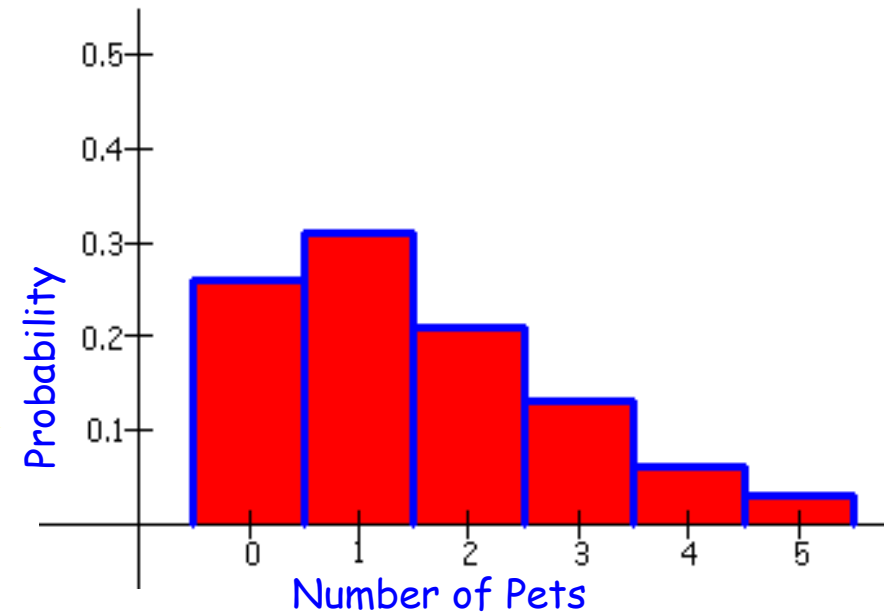
Then, the Department of Animal Control has collected data over the course of several years. They want to estimate the long-run probabilities for the values of x .



- The results as follows:

x	0	1	2	3	4	5
$p(x)$	0.26	0.31	0.21	0.13	0.06	0.03

This is called a **discrete probability** distribution. It can also be displayed in a histogram with the probability on the vertical axis.



Properties of Discrete Probability Distributions

1) For every possible x value,

$$0 \leq p(x) \leq 1.$$

2) $\sum_{\text{all } x \text{ values}} p(x) = 1$

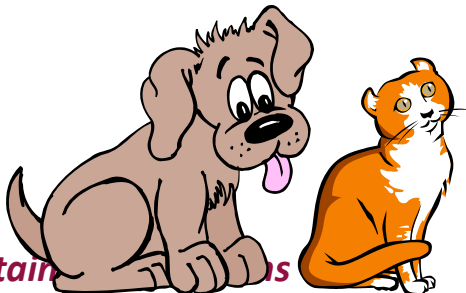
Example

Refer to previous example (Dogs and Cats) .

x	0	1	2	3	4	5
$p(x)$	0.26	0.31	0.21	0.13	0.06	0.03

What is the probability that a randomly selected household in Wolf City has at most 2 pets?

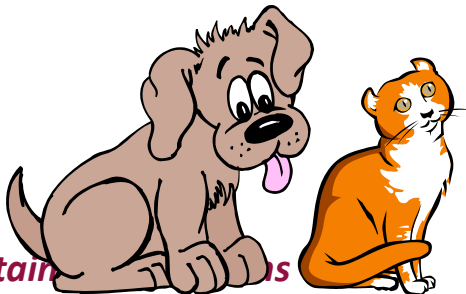
$$p(x \leq 2) = 0.26 + 0.31 + 0.21 = 0.78$$



x	0	1	2	3	4	5
$p(x)$	0.26	0.31	0.21	0.13	0.06	0.03

What is the probability that a randomly selected household in Wolf City has less than 2 pets?

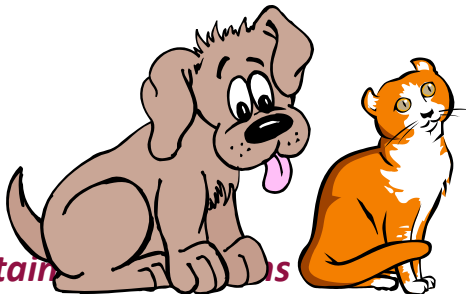
$$p(x < 2) = 0.26 + 0.31 = 0.57$$



x	0	1	2	3	4	5
$p(x)$	0.26	0.31	0.21	0.13	0.06	0.03

What is the probability that a randomly selected household in Wolf City has more than 1 but no more than 4 pets?

$$p(1 < x \leq 4) = 0.21 + 0.13 + 0.06 = 0.40$$



Exercise #1

- Let y denote the number of broken eggs in a randomly selected carton of one dozen eggs. Suppose that the probability of y is as follows:

y	0	1	2	3	4
$P(y)$	0.65	0.2	0.1	0.04	?

- What is $p(4)$
- Calculate $p(y \leq 2)$
- The probability that the carton contains fewer than two broken eggs

Exercise #2

- Ahmad volunteers in community events each month. He does not do more than five events in a month. He attends exactly five events 35% of the time, four events 25% of the time, three events 20% of the time, two events 10% of the time, one event 5% of the time, and no events 5% of the time.
 - a) Define the random variable X ?
 - b) What values does x take on?
 - c) Find the probability that Ahmad volunteers for less than three events each month.
 - d) Find the probability that Ahmad volunteers for at least one event each month.

Solution

- X = the number of event ahmad volunteer for each month
- Value of $X = \{0, 1, 2, 3, 4, 5\}$

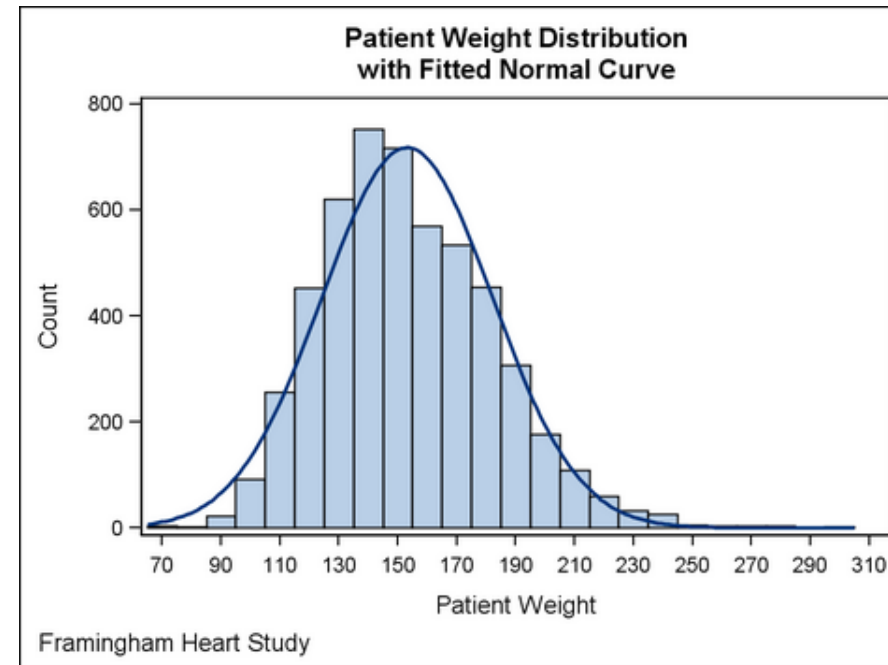
X	0	1	2	3	4	5
$P(x)$	0.05	0.05	0.10	0.20	0.25	0.35

- $P(x < 3) = P(x=0) + P(x=1) + P(x=2) = 0.05 + 0.05 + 0.10 = 0.2$
- $P(x > 0) = 1 - 0.05 = 0.95$

Probability Distribution For CONTINUOUS Variables

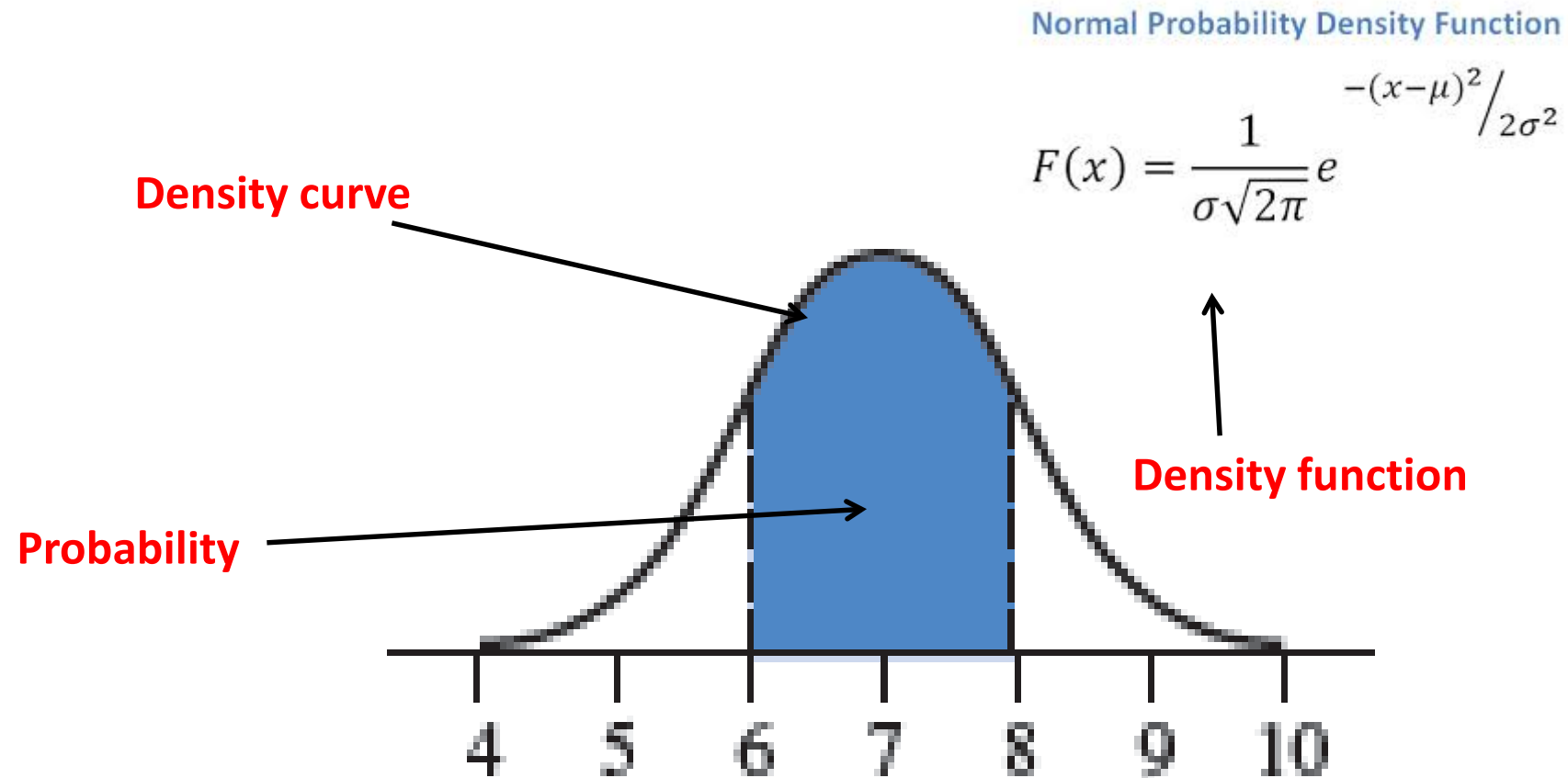
Definition : Continuous Random Variable

- A continuous random variables is a random variable where the data can take infinitely many values.
- A continuous random variable “ x ” takes all values in a given interval of numbers.



Definition : Probability Distribution for Continuous Random Variable

- It is specified by a curve called a **density curve**.
- The function that describes this curve is denoted by $f(x)$ and is called the **density function**.
- The probability of observing a value in a particular interval is the **area under the curve** and above the given interval.



Properties of Continuous Probability Distributions

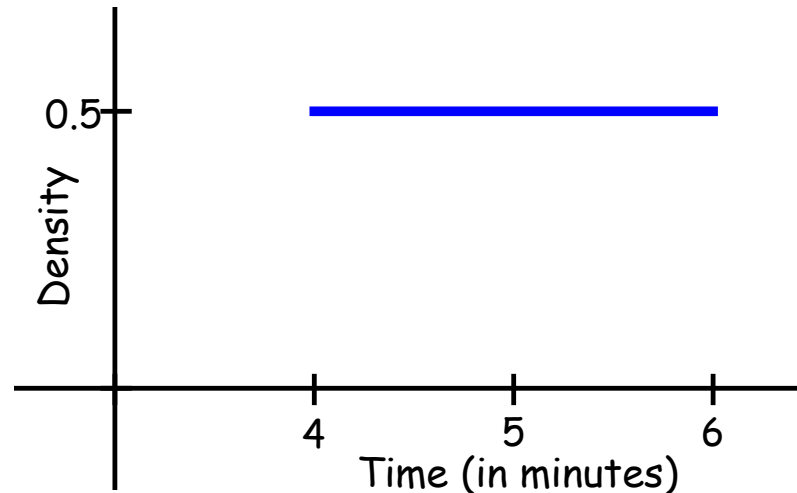
1. $f(x) \geq 0$ (the curve cannot dip **below** the horizontal axis).
2. The **total area** under the density curve equals **one**.

Example

Suppose x is a continuous random variable defined as the amount of time (in minutes) taken by a clerk to process a certain type of application form. Suppose x has a probability distribution with density function:

$$f(x) = \begin{cases} .5 & 4 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

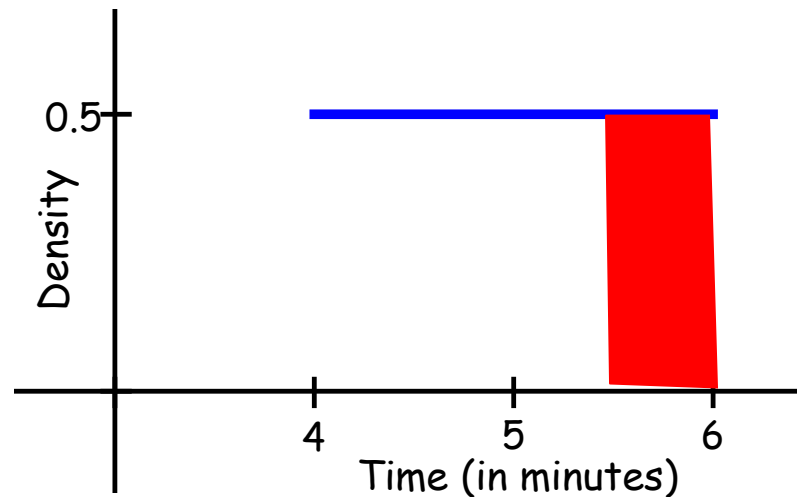
The following is the graph of $f(x)$, the density curve:



Example (cont.)

What is the probability that it takes more than 5.5 minutes to process the application form?

$$p(x > 5.5) = 0.5(0.5) = 0.25$$



Other Density Curves

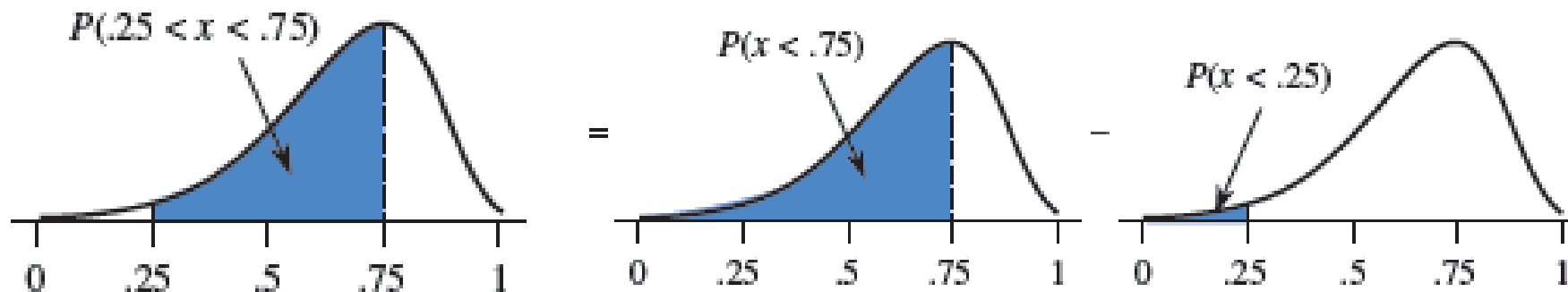
Some density curves resemble the one below. Integral calculus is used to find the area under these curves.

We will use tables (with the values already calculated).

The probability that a continuous random variable x lies between a **lower limit** a and an **upper limit** b is

$$p(a < x < b) = (\text{cumulative area to the left of } b) - (\text{cumulative area to the left of } a)$$

$$p(a < x < b) = p(x < b) - p(x < a)$$



- Other continuous distribution

We will use tables for certain distribution (with the values already calculated).

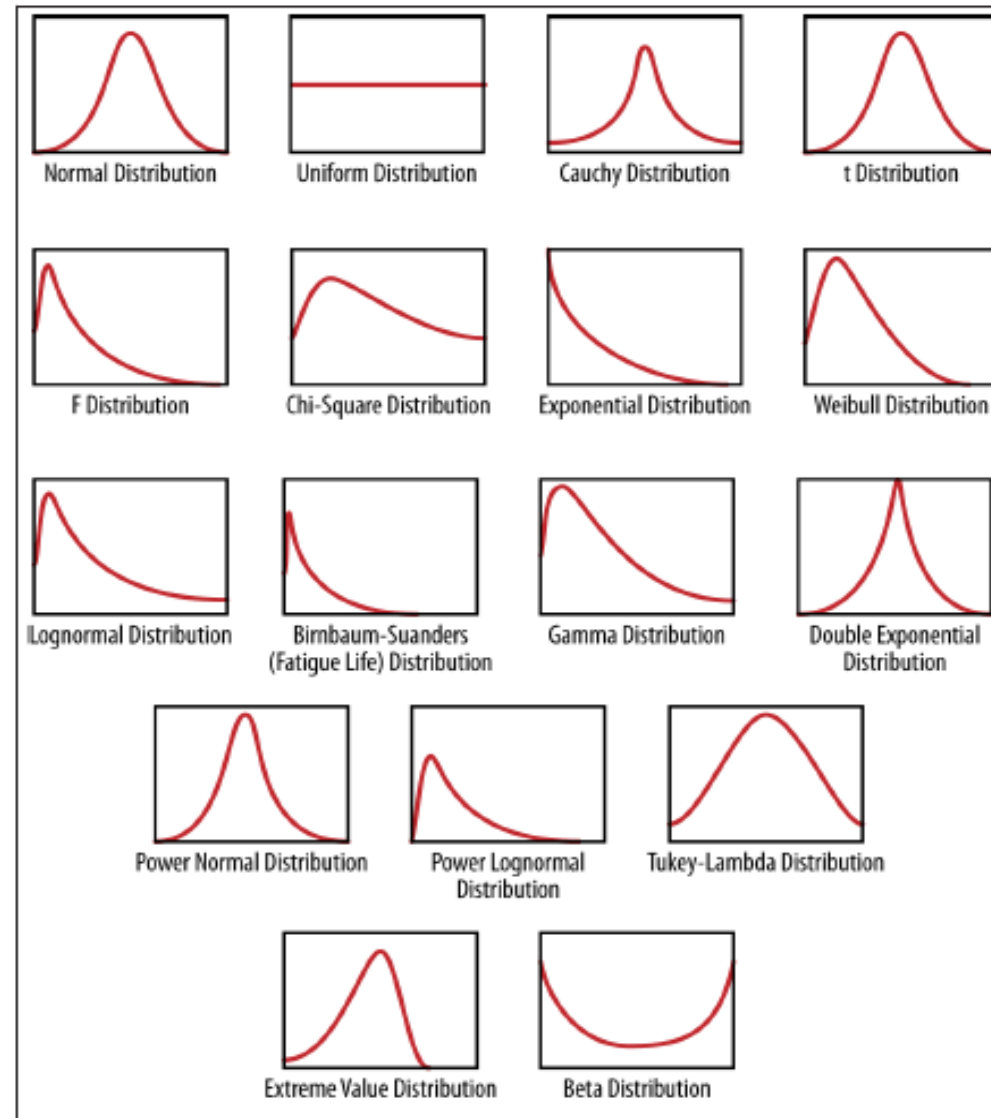


Figure 2-1. A bunch of continuous density functions (aka probability distributions)

Discrete & Continuous Probability Distribution

Special Distributions

Discrete Distribution:

- Binomial
- Negative Binomial
- Geometric

Continuous Distribution:

- Normal

Discrete Distribution

Binomial Distribution

- An experiments often consists of repeated trials, each with two possible outcomes that may be labeled **success** or **failure**.
- **Example**: The testing items as they come off an assembly line, where each trial may indicate a defective or a non-defective item.
- The process is referred to as a **Bernoulli process**.
- Each trial is called a **Bernoulli trial**.

Binomial Distribution

Suppose we decide to record the gender of the next 25 newborns at a particular hospital.

What is the chance that at least 15 are female?



What is the chance that between 10 and 15 are female?

Out of the 25 newborns, how many can we expect to be female?

Binomial Distribution (cont'd)

Properties of a Bernoulli process:

1. There are a **fixed number** of trials (can be decide in advance).
2. Each trial results in **one of only two possible** outcomes, labeled success (S) or failure (F).
3. Outcomes of different trials are **independent**.
4. The probability that a trial results in success is the **same** for all trials.

Binomial Distribution (cont'd)

- The number X of success in n Bernoulli trials is called a binomial random variable.
- The probability distribution of this discrete random variable is called the **binomial distribution**.

- **Formula:**

A Bernoulli trial can result in a success with probability p and a failure with probability $q = 1-p$. Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

- The **mean** and **variance** of the binomial distribution:

$$\mu = np ; \sigma^2 = npq ; \sigma = \sqrt{npq}$$

How to pick variable in binomial distribution

Approximately 70% of statistics students do their homework in time for it to be collected and graded. Each student does homework independently. In a statistics class of 50 students, what is the probability that at least 40 will do their homework on time? Students are selected randomly.

- a) Is this binomial problem and why?
- b) If we are interested in the number of students who do their homework on time, then how do we define X ? What values does take on?
- c) What is a "failure," in words?
- d) If $p+q=1$, then what is q ?
- e) What kind of *inequality* used to translate the words "at least". E.g $P(x \geq 40)$

Answer

- a) A binomial problem because there is only a success or a failure, and a fixed number of trials with probability of a success is 0.70 for each trial.
- b) X = the number of statistics students who do their homework on time
- c) 0, 1, 2, ..., 50.
- d) Failure is defined as a student who does not complete his or her homework on time. The probability of a success is $p=0.7$. The number of trial is $n=50$.
- e) $q=0.3$
- f) $p(x \geq 40)$

Example

The probability that a certain kind of component will survive a shock test is $\frac{3}{4}$. Find the probability that exactly 2 of the next 4 components tested survive.

Solution:

Assuming that the tests are independent and $p = \frac{3}{4}$ for each of the 4 tests, we obtain

$$b(2; 4, \frac{3}{4}) = \binom{4}{2} \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 = \frac{27}{128}$$

Example

A coin is tossed four times. Find the **mean** and **variance** of the number of heads that will be obtained? Given the distribution as follows.

No of head, x	0	1	2	3	4
Probability	1/16	4/16	6/16	4/16	1/16

Solution:

$$n = 4, p = \frac{1}{2}; q = \frac{1}{2}$$

$$\therefore \mu = 4 \left(\frac{1}{2} \right) = 2;$$

$$\therefore \sigma^2 = 4 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 1$$

Negative Binomial Distribution

If repeated independent trials can result in a success with probability p and a failure with probability $q = 1 - p$, then the probability distribution of the random variable x , **the number of the trial** on which the **k -th** success occurs, is

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}, x = k, k+1, k+2, \dots$$

Example

A scientist inoculates mice, one at a time, with a diseases germ until he finds 2 that have contracted the diseases. If the probability of contracting the diseases is $1/6$, what is the probability that 8 mice are required?

Solution:

$$b^*(8; 2, 0.1667) = \binom{8-1}{2-1} (0.1667)^2 (1 - 0.1667)^{8-2} = 0.0651$$

Exercise #4

In an NBA (National Basketball Association) championship series, the team that wins four games out of seven is the winner. Suppose that teams A and B face each other in the championship games and that team A has probability 0.55 of winning a game over team B.

- a) What is the probability that team A will win the series in 6 games?
- b) What is the probability that team A will win the series?
- c) If teams A and B were facing each other in a regional playoff series, which is decided by winning three out of five games, what is the probability that team A would win the series?

Geometric Distribution

Properties of Geometric Distributions:

1. The trial is **independent**.
2. Each trial can result in **one of two possible** outcomes, success and failure.
3. The probability of success is the **same** for all trials.

A geometric random variable x is defined as

x = number of trials UNTIL the FIRST success is observed (including the success trial).

Geometric Distribution (cont.)

Geometric distribution is in a sense **INFINITE**. They're asking you what is the probability **UNTIL** the first success happens.

Example:

Malik throws a dice until he gets a '4'.

What is the probability of him getting the first '4' in the 6th throw? How to calculate the probability?

Geometric Distribution (cont.)

Geometric Probability Distribution function:

If x is a random variable with probability of success = p for each trial, then

$$p(x) = (1 - p)^{x-1} p, \quad x = 1, 2, 3, \dots$$

Geometric Distribution (cont.)

Formulas for mean and standard deviation of a geometric distribution:

$$\text{mean: } \mu_x = \frac{1}{p}$$

$$\text{std. dev: } \sigma_x = \sqrt{\frac{(1-p)}{p^2}}$$

Example

A product produced by a machine has a 3% defective rate. What is the probability that the first defective occurs in the fifth(5th) item inspected?

Solution:

$$\begin{aligned} p(x = 5) &= (1 - 0.03)^{5-1} (0.03) \\ &= 0.027 \end{aligned}$$

Exercise #5

It is claimed that 15% of the ducks in a particular region have patent schistosome infection. Suppose that seven ducks are selected at random. Let x equals the number of ducks that are infected.

- a) Assuming independence trials, how is x distributed?
- b) Find the probability that:
 - i) At most two ducks are infected.
 - ii) One duck is not infected.
- c) A new drug is proposed to treat the schistosome infection. Therefore, the infected duck is randomly selected. What is the probability that the first duck infected is found at the fifth trial?

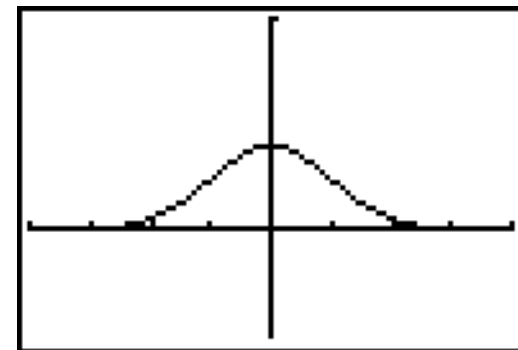
Exercise #6

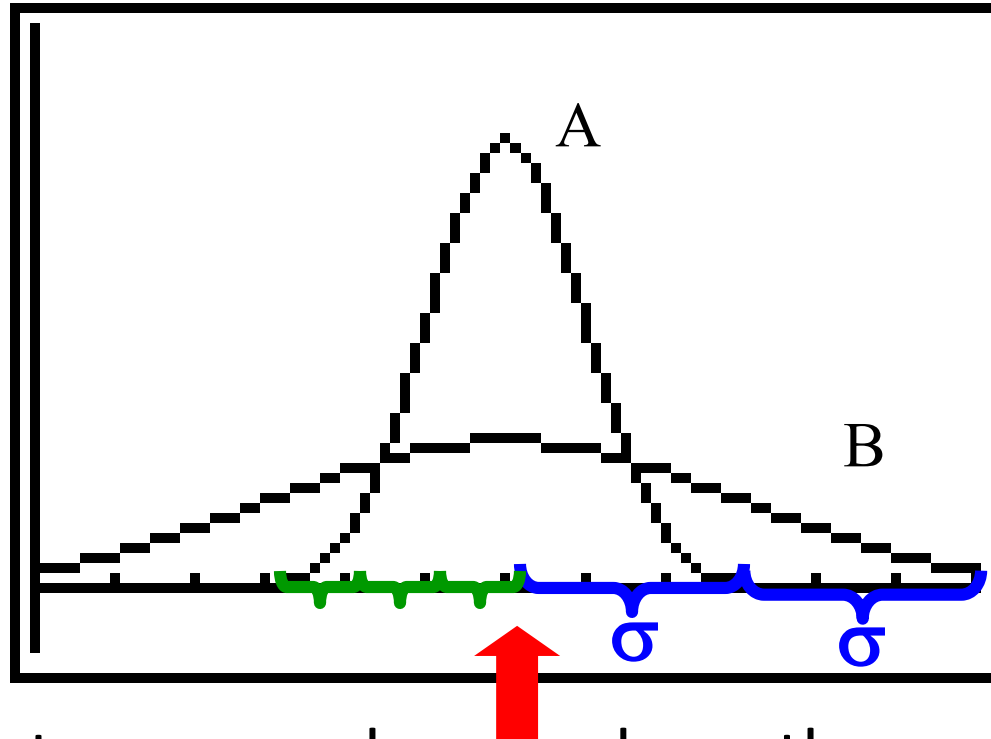
A company fabricates special-purpose robots and records show that the probability is 0.10 that the one of its new robots will require repairs during confirmation tests. What is the probability that the eighth robot it builds in a month is the first one to require repairs?

Continuous Distribution

Continuous Distribution

- Symmetrical bell-shaped (unimodal) density curve defined by m and s
- Area under the curve equals 1.
- Probability of observing a value in a particular interval is calculated by finding the area under the curve.
- As s increases, the curve flattens & spreads out.
- As s decreases, the curve gets taller and thinner.



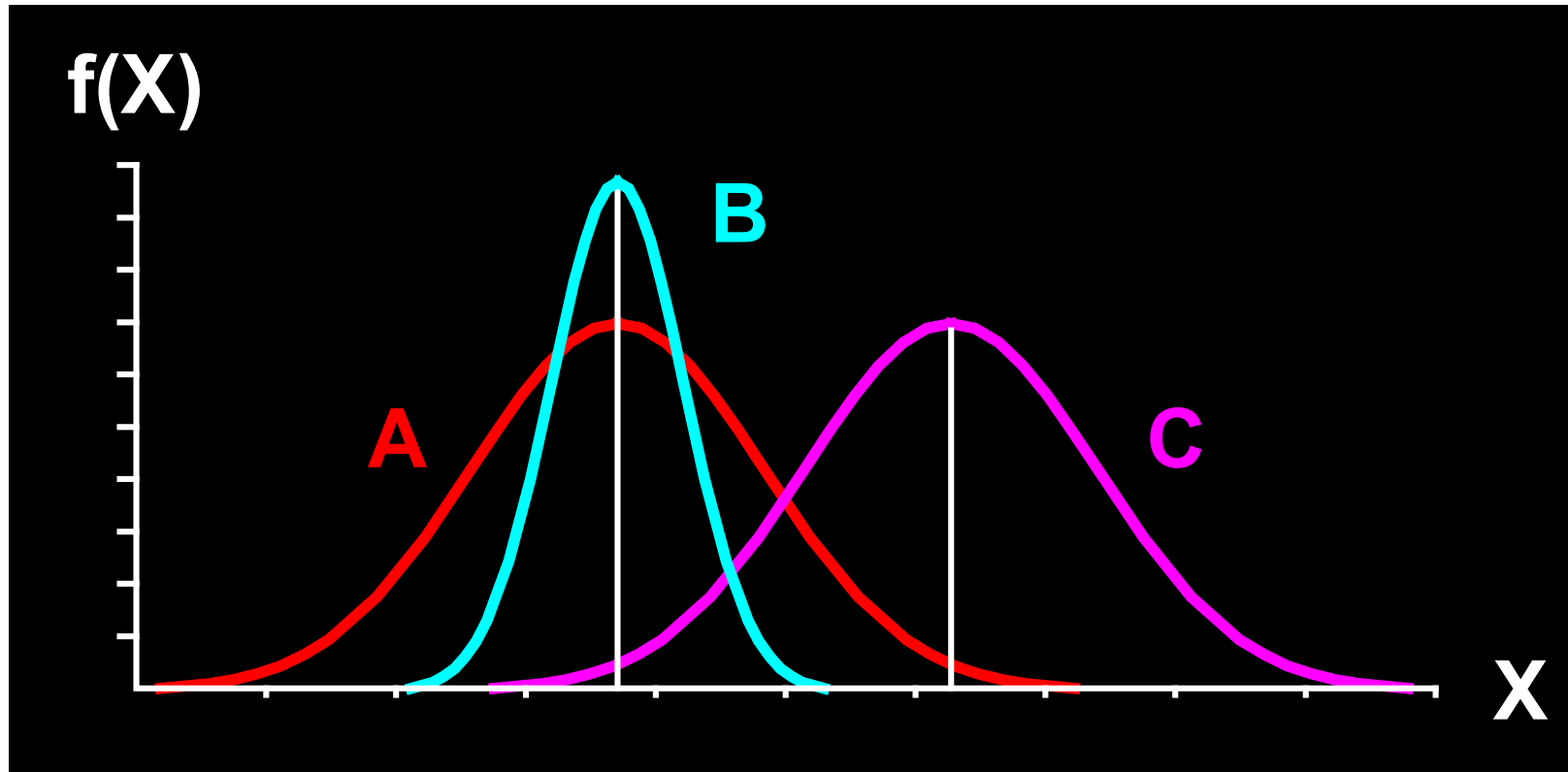


Do these two normal curves have the same mean? If so, what is it? **YES**

Which normal curve has a standard deviation of 3? **B**

Which normal curve has a standard deviation of 1? **A**

Effect of Varying Parameters (μ & σ)



Notation

x is $N(\mu, \sigma)$

The random variable x has a normal distribution (N) with mean, μ and standard deviation, σ .

Example:

x is $N(40,1)$; $x \sim N(40,1)$

x is $N(10,5)$; $x \sim N(10,5)$

x is $N(50,3)$; $x \sim N(50,3)$

Probability Density Function

Normal Distribution

Parameters: μ and σ

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x - \mu)^2}{2\sigma^2}\right\}$$

for $-\infty < x < \infty$

x = value of random variable ($-\infty < x < +\infty$)

σ = Population Standard Deviation

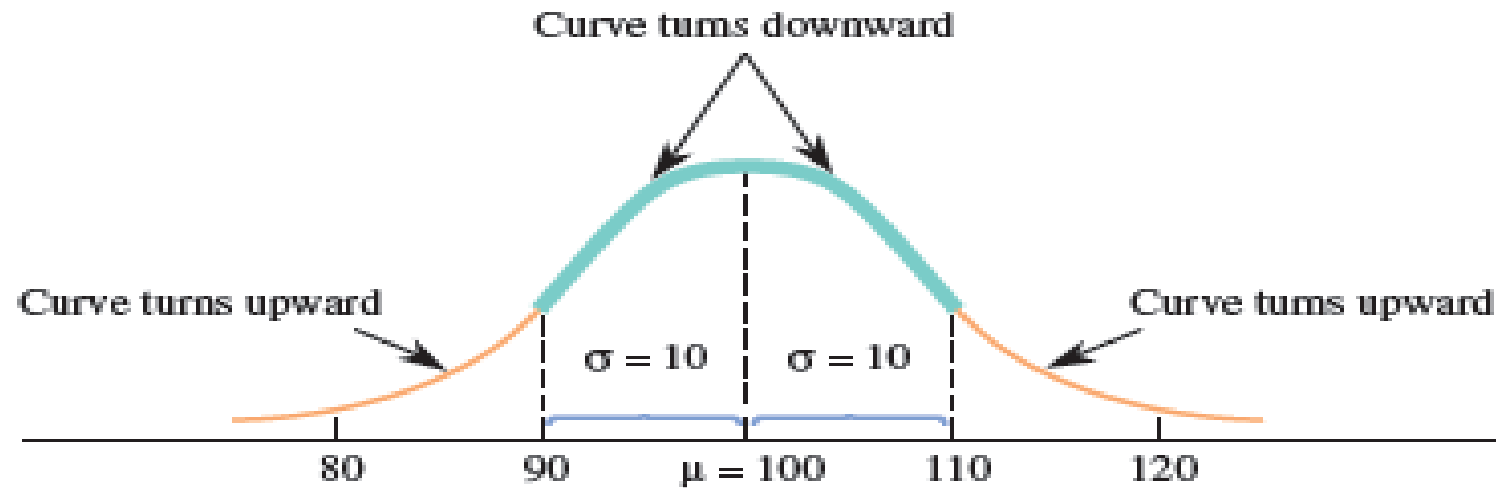
π = 3.14159

e = 2.71828

μ = Mean of Random Variable x

Normal Curve

Notice that the normal curve is curving downwards from the center (mean) to points that are one standard deviation on either side of the mean. At those points, the normal curve begins to turn upward.



Standard Normal Distribution

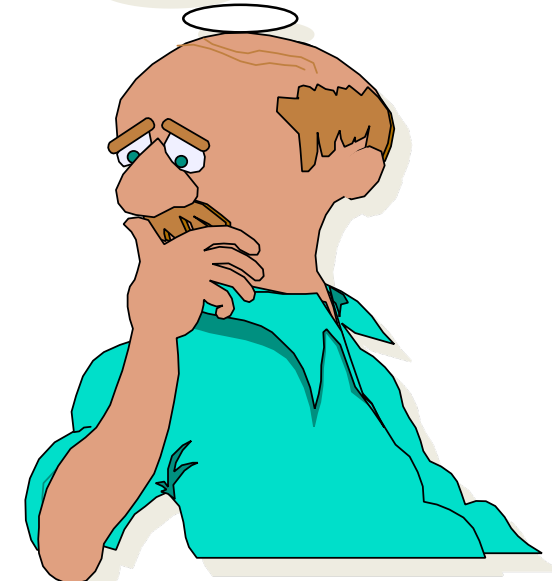
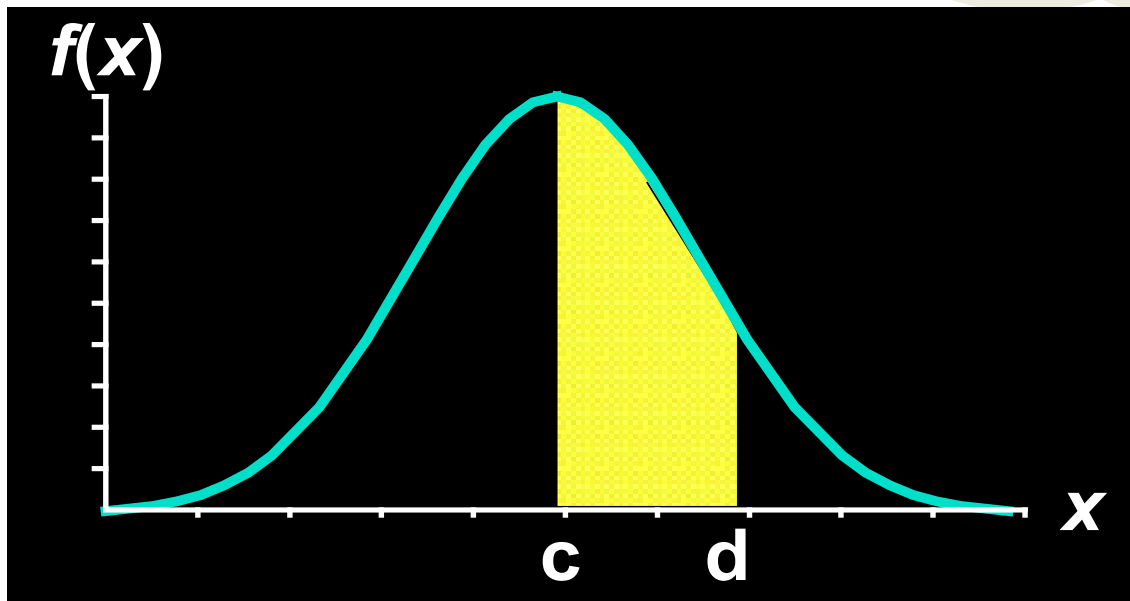
- Is a normal distribution with $\mu = 0$ and $\sigma = 1$
- It is customary to use the letter z to represent a variable whose distribution is described by the standard normal curve (or z curve).

▪

Normal Distribution Probability

Probability is
area under
curve!

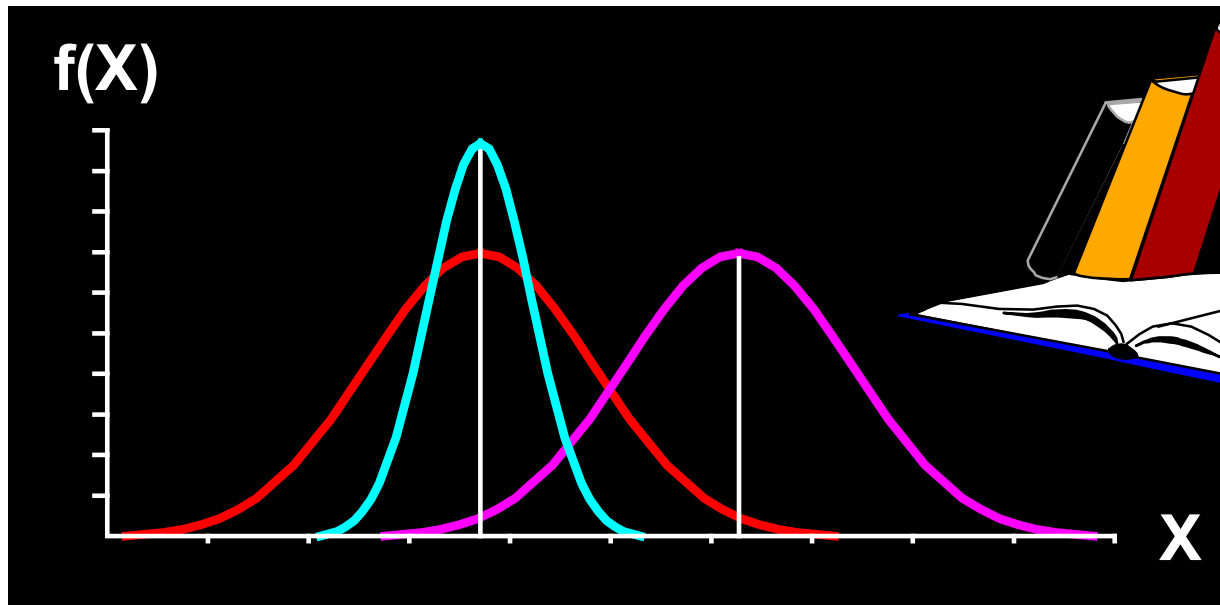
$$P(c \leq x \leq d) = \int_c^d f(x) dx ?$$



Infinite Number of Distribution Tables

Normal distributions differ by
mean & standard deviation.

Each distribution would
require its own table.



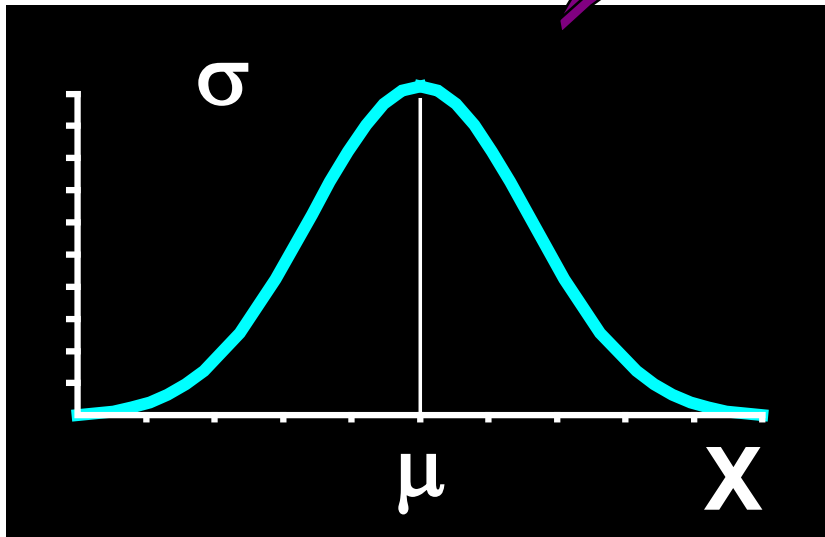
That's an *infinite* number!

Standardize the Normal Distribution

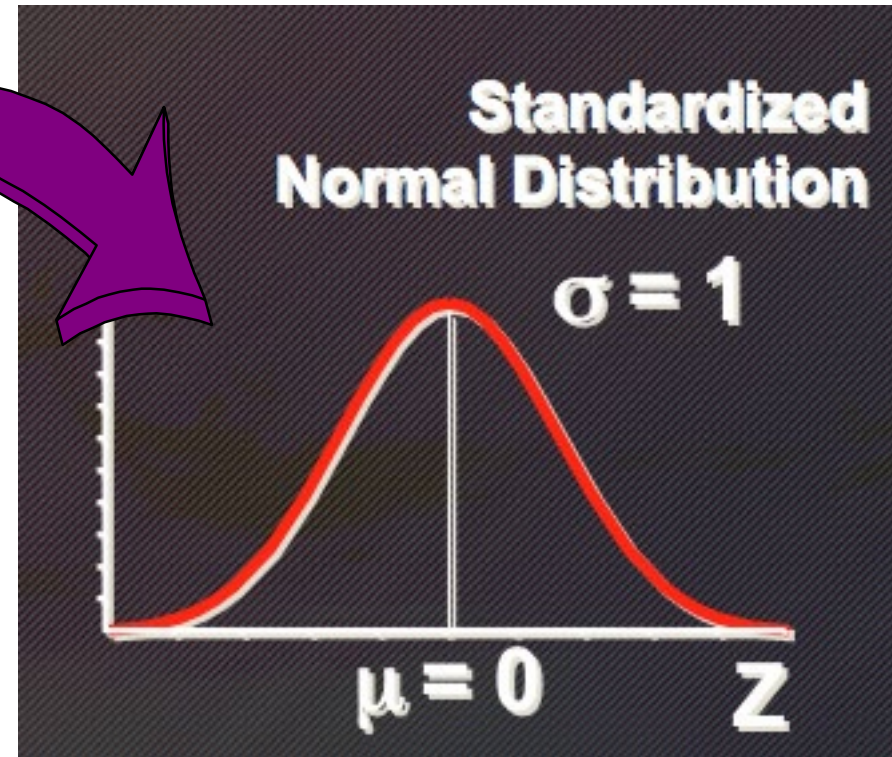
$$Z = \frac{X - \mu}{\sigma}$$

$$x \sim N(0, 1)$$

Normal
Distribution



Standardized
Normal Distribution



One table!

There are 2 type of Z distribution tables:

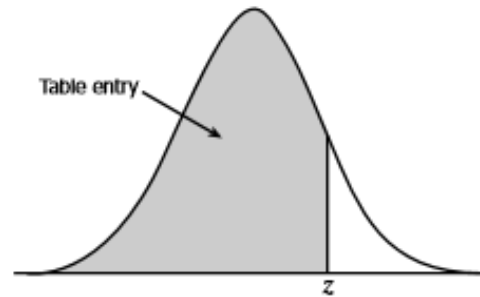


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141

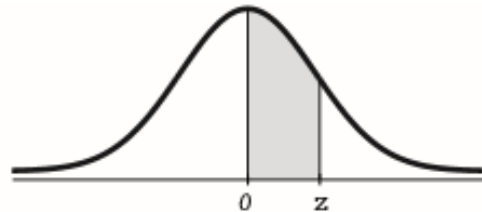


TABLE A-2 Standard Normal (z) Distribution										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141

Full Z table

The probability area is calculated from the very left area

Half Z table

The probability area is calculated from the center area

For full Z distribution tables, there are two different tables representing the area:

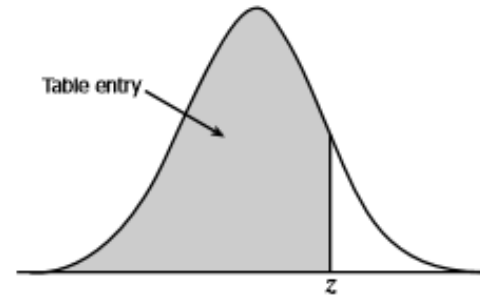


Table entry for z is the area under the standard normal curve to the left of z .

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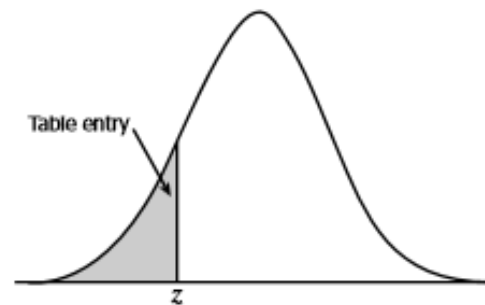


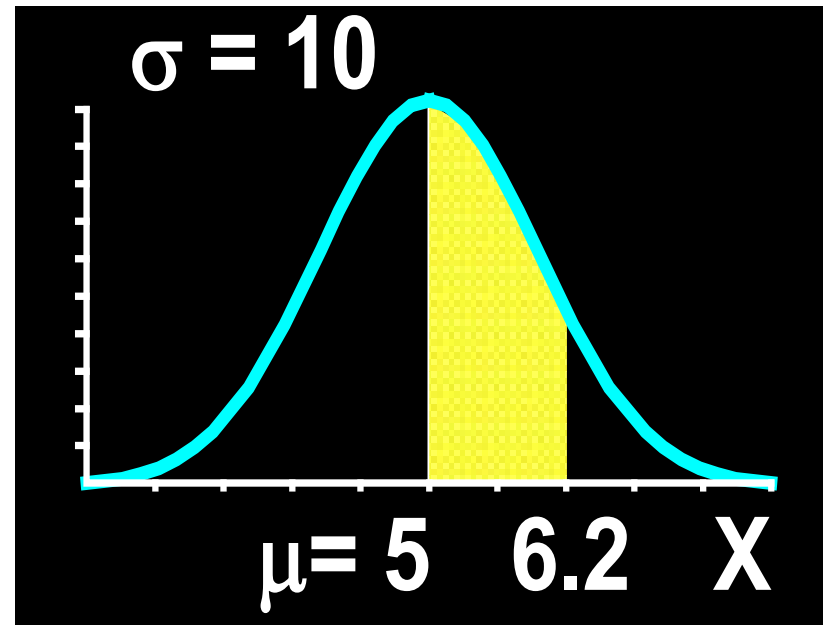
Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005

Example

What is the probability of $P(X=6.2)$, given that the mean is 5 and std. deviation is 10?.

Normal Distribution

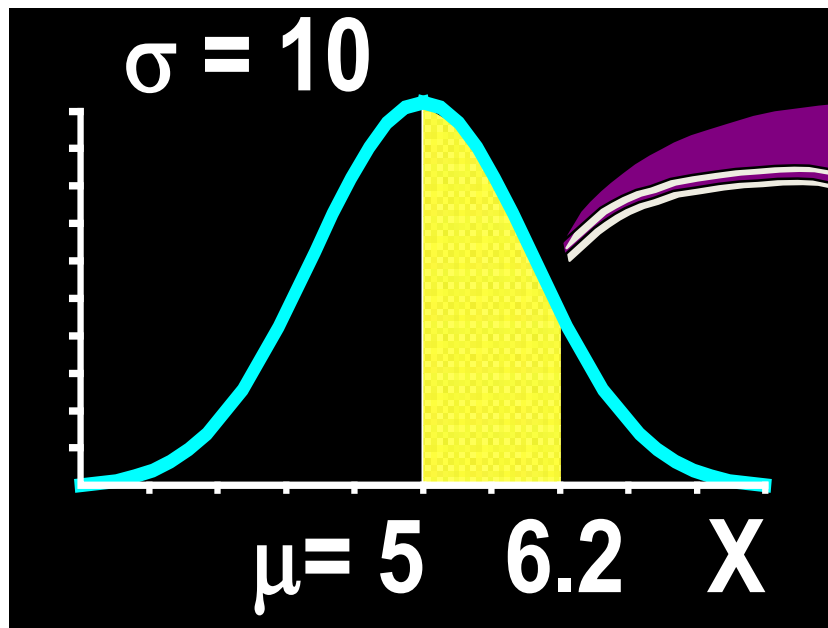


Example -Solution

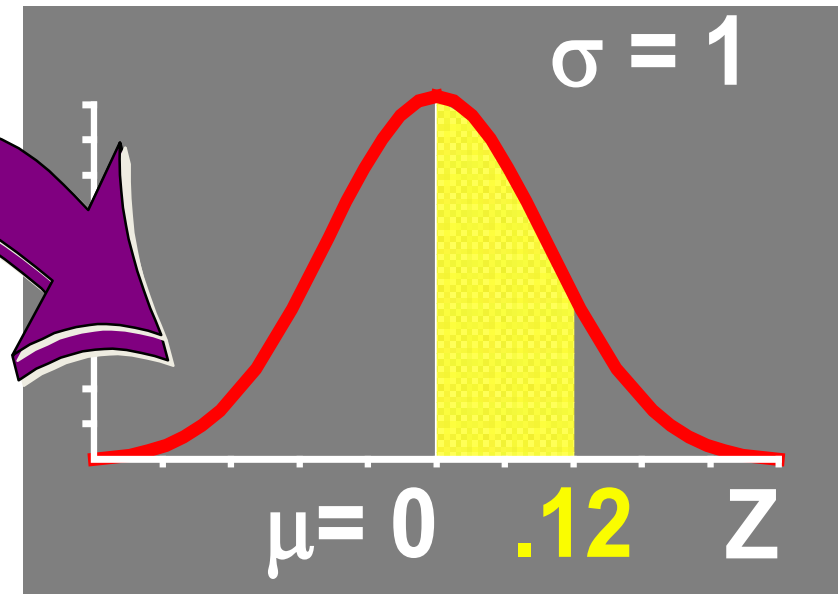
Step 1: Find the Z value.

$$Z = \frac{X - \mu}{\sigma} = \frac{6.2 - 5}{10} = 0.12$$

Normal Distribution



Standardize Normal Distribution



Note: Shaded area exaggerated.

Step 2: Refer to standard Normal Dist. Table to locate the Z value.

Standardized Normal
Probability Table (Portion)

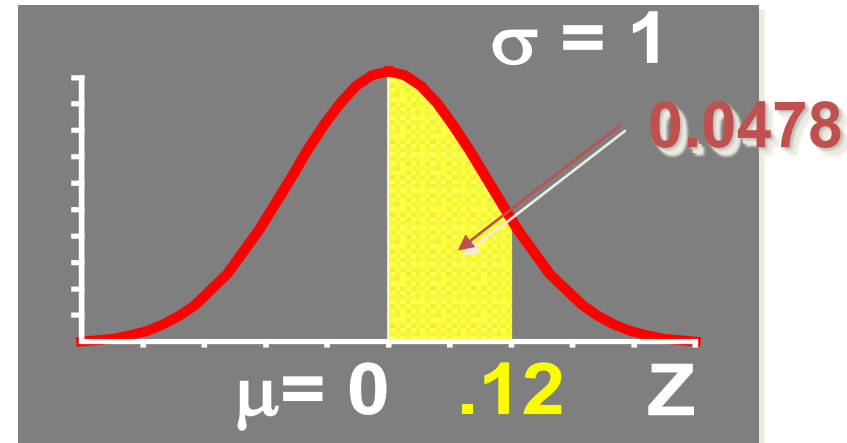


Full Z Table

z	.00	.01	.02	.03	.04
0.0	.5000	.5040	.5080	.5120	.5160
0.1	.5398	.5438	.5478	.5517	.5557
0.2	.5793	.5832	.5871	.5910	.5948
0.3	.6179	.6217	.6255	.6293	.6331
0.4	.6554	.6591	.6628	.6664	.6700

Half Z Table

z	.00	.01	.02	.03
0.0	.0000	.0040	.0080	.0120
0.1	.0398	.0438	.0478	.0517
0.2	.0793	.0832	.0871	.0910
0.3	.1179	.1217	.1255	.1293
0.4	.1554	.1591	.1628	.1664



$$P(Z = 0.12) = 0.5478 - 0.5 = 0.0478$$

Example: Find $P(3.8 \leq X \leq 5)$

$$Z = \frac{X - \mu}{\sigma} = \frac{3.8 - 5}{10} = -0.12$$

Full Z Table

z	.00	.01	.02	.03
-3.4	.0003	.0003	.0003	.0003
-3.3	.0005	.0005	.0005	.0004
...	...	...	...	...
-0.2	.4207	.4168	.4129	.4090
-0.1	.4602	.4562	.4522	.4483
-0.0	.5000	.4960	.4920	.4880

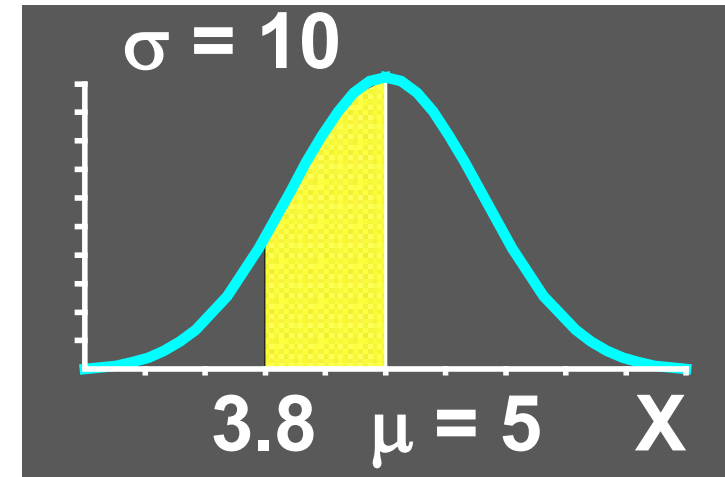
$$P(Z=0.12) = 0.5 - 0.4522$$

$$= 0.0478$$

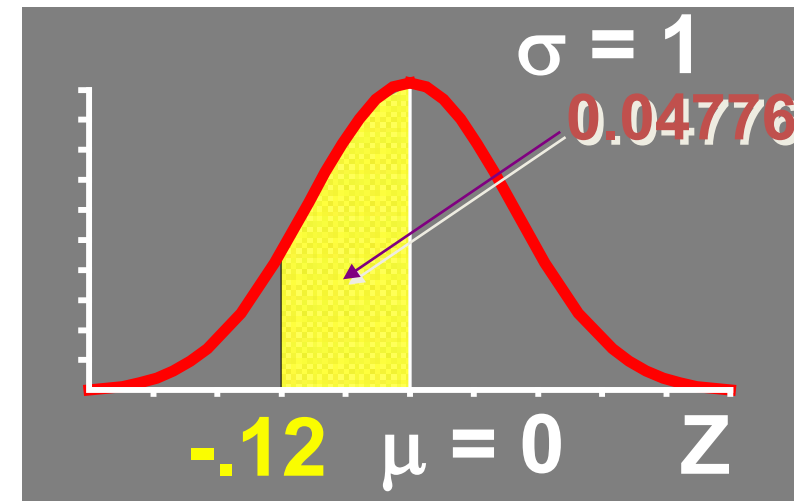
Half Z Table

z	.00	.01	.02	.03
0.0	.0000	.0040	.0080	.0120
0.1	.0398	.0438	.0478	.0517
0.2	.0793	.0832	.0871	.0910
0.3	.1179	.1217	.1255	.1293
0.4	.1554	.1591	.1628	.1664

Normal Distribution

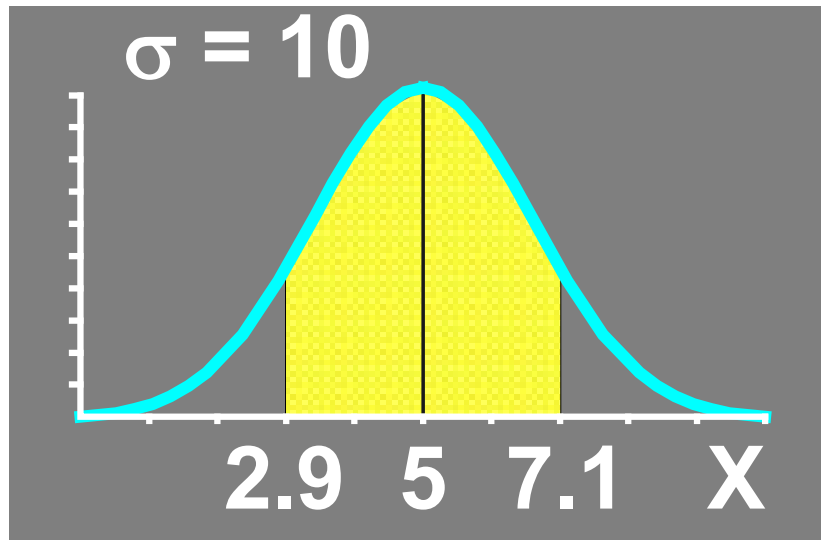


Standardize Normal Distribution



Example: Find $P(2.9 \leq X \leq 7.1)$

Normal Distribution



$$Z = \frac{X - \mu}{\sigma} = \frac{2.9 - 5}{10} = -.21$$

$$Z = \frac{X - \mu}{\sigma} = \frac{7.1 - 5}{10} = .21$$

Example $P(2.9 \leq X \leq 7.1)$

$$Z = \frac{X - \mu}{\sigma} = \frac{2.9 - 5}{10} = -0.21$$

Full Z Table

z	.00	.01	.02	.03
-3.4	.0003	.0003	.0003	.0003
-3.3	.0005	.0005	.0005	.0004
...	...	...	...	...
-0.2	.4207	.4168	.4129	.4090
-0.1	.4602	.4562	.4522	.4483
-0.0	.5000	.4960	.4920	.4880

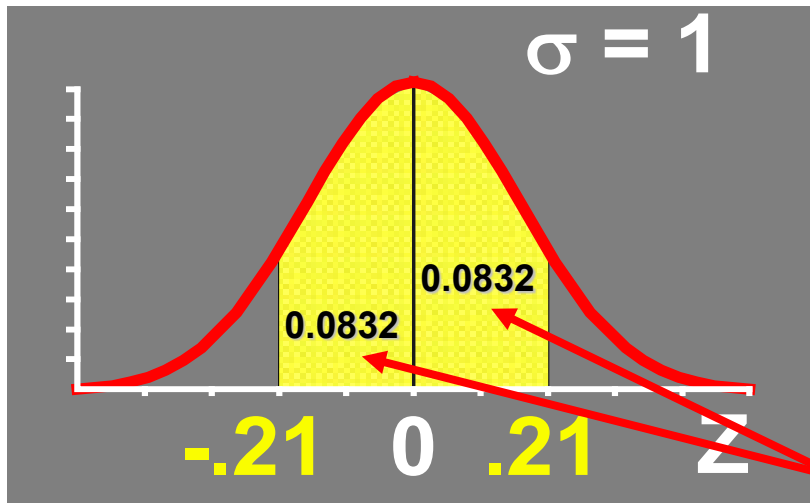
$$P(Z = -0.21) = 0.5 - 0.4168 = 0.0832$$

$$Z = \frac{X - \mu}{\sigma} = \frac{7.1 - 5}{10} = 0.21$$

Full Z Table

z	.00	.01	.02	.03	.04
0.0	.5000	.5040	.5080	.5120	.5160
0.1	.5398	.5438	.5478	.5517	.5557
-0.2	.5793	.5832	.5871	.5910	.5948
0.3	.6179	.6217	.6255	.6293	.6331
0.4	.6554	.6591	.6628	.6664	.6700

$$P(Z = 0.21) = 0.5832 - 0.5 = 0.0832$$



Half Z Table

z	.00	.01	.02	.03
0.0	.0000	.0040	.0080	.0120
0.1	.0398	.0438	.0478	.0517
-0.2	.0793	.0832	.0871	.0910
0.3	.1179	.1217	.1255	.1293
0.4	.1554	.1591	.1628	.1664

$$0.0832 + 0.0832 = 0.1664$$

Example: Find $P(X \geq 8)$

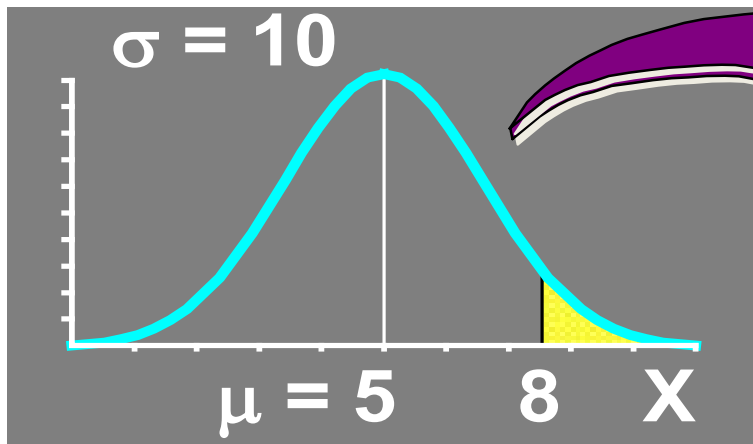
$$Z = \frac{X - \mu}{\sigma} = \frac{8 - 5}{10} = .30$$

Full Z Table

z	.00	.01	.02	.03	.04
0.0	.5000	.5040	.5080	.5120	.5160
0.1	.5398	.5438	.5478	.5517	.5557
0.2	.5793	.5832	.5871	.5910	.5948
-0.3 →	.6179	.6217	.6255	.6293	.6331
0.4	.6554	.6591	.6628	.6664	.6700

$$P(Z = 0.30) = 1 - 0.6179 = 0.3821$$

Normal Distribution

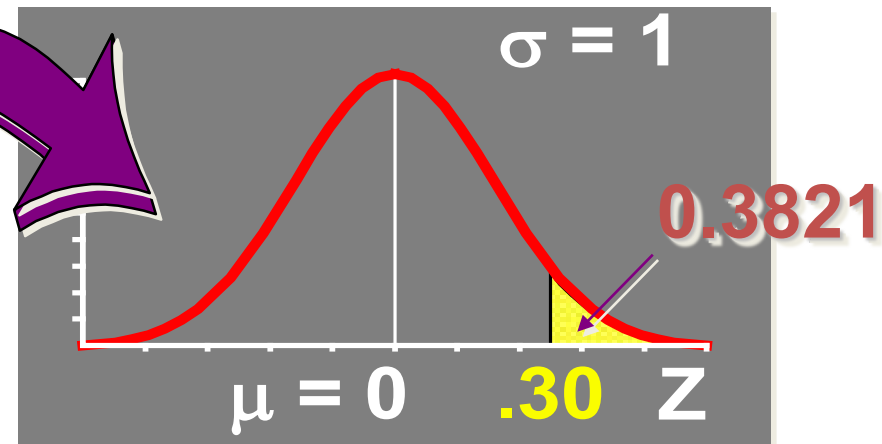


Half Z Table

z	.00	.01	.02	.03
0.0	.0000	.0040	.0080	.0120
0.1	.0398	.0438	.0478	.0517
0.2	.0793	.0832	.0871	.0910
-0.3 →	.1179	.1217	.1255	.1293
0.4	.1554	.1591	.1628	.1664

$$P(Z = 0.30) = 0.5 - 0.1179 = 0.3821$$

Standardize Normal Distribution



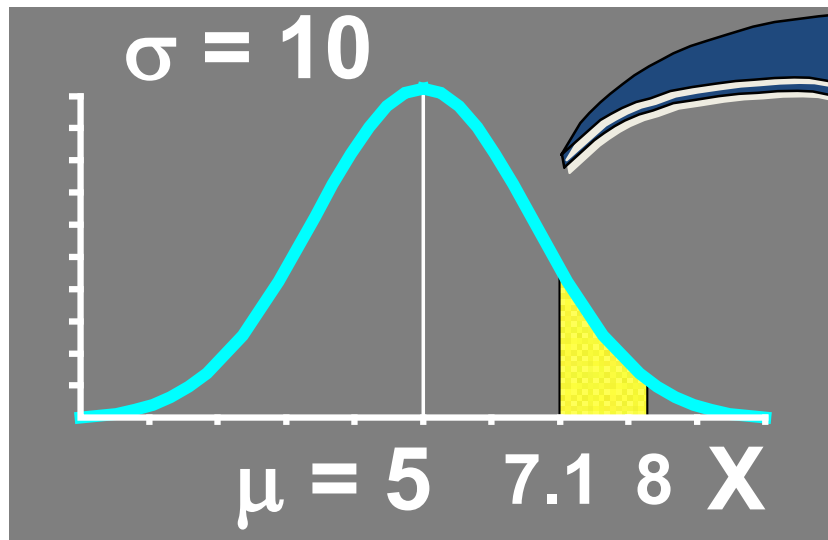
Example: Find $P(7.1 \leq X \leq 8)$

$$Z = \frac{X - \mu}{\sigma} = \frac{7.1 - 5}{10} = 0.21 \quad \Rightarrow \quad P(Z = 0.21) = 0.0832$$

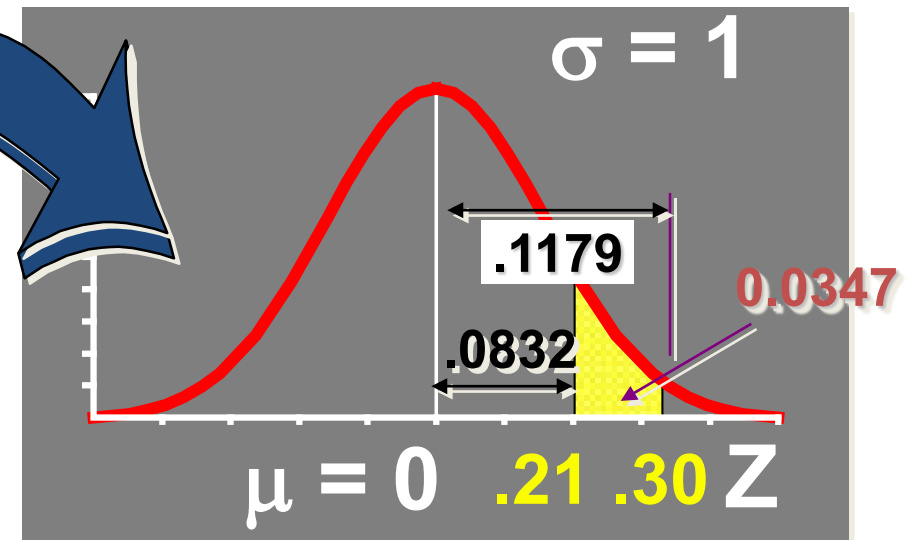
$$Z = \frac{X - \mu}{\sigma} = \frac{8 - 5}{10} = 0.30 \quad \Rightarrow \quad P(Z = 0.30) = 0.1179$$

$$P(0.21 \leq Z \leq 0.30) = 0.1179 - 0.0832 = 0.0347$$

Normal Distribution



Standardize Normal Distribution

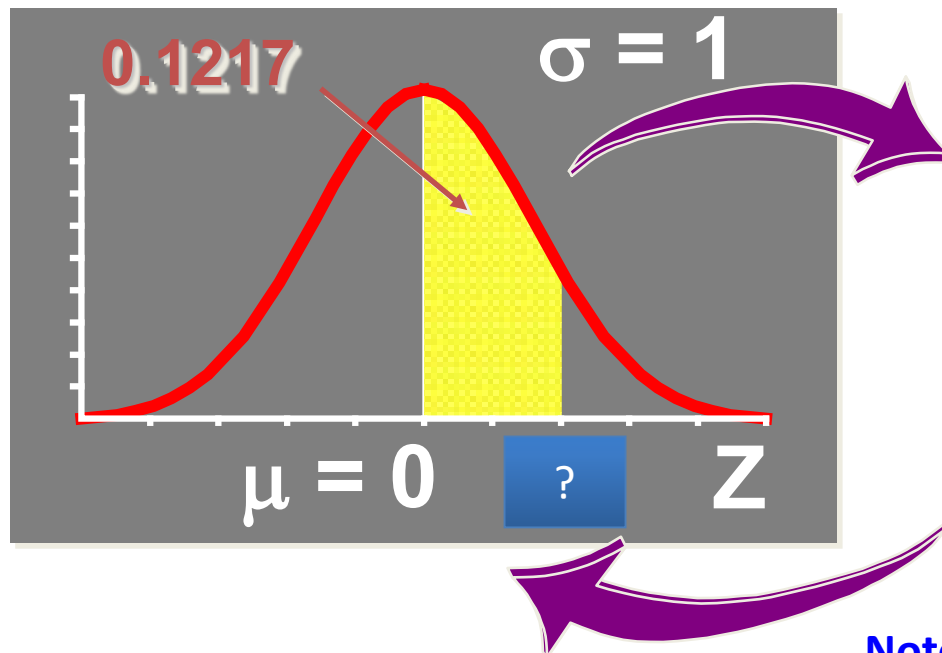


Finding Z Values for Known Probabilities

If Z greater than mean, what is Z given that $P(Z) = 0.1217$?

$$P(Z=?) = a - 0.5 = 0.1217 \Rightarrow 0.1217 + 0.5 = 0.6217 \text{ Full Z Table}$$

Standardize Normal Distribution



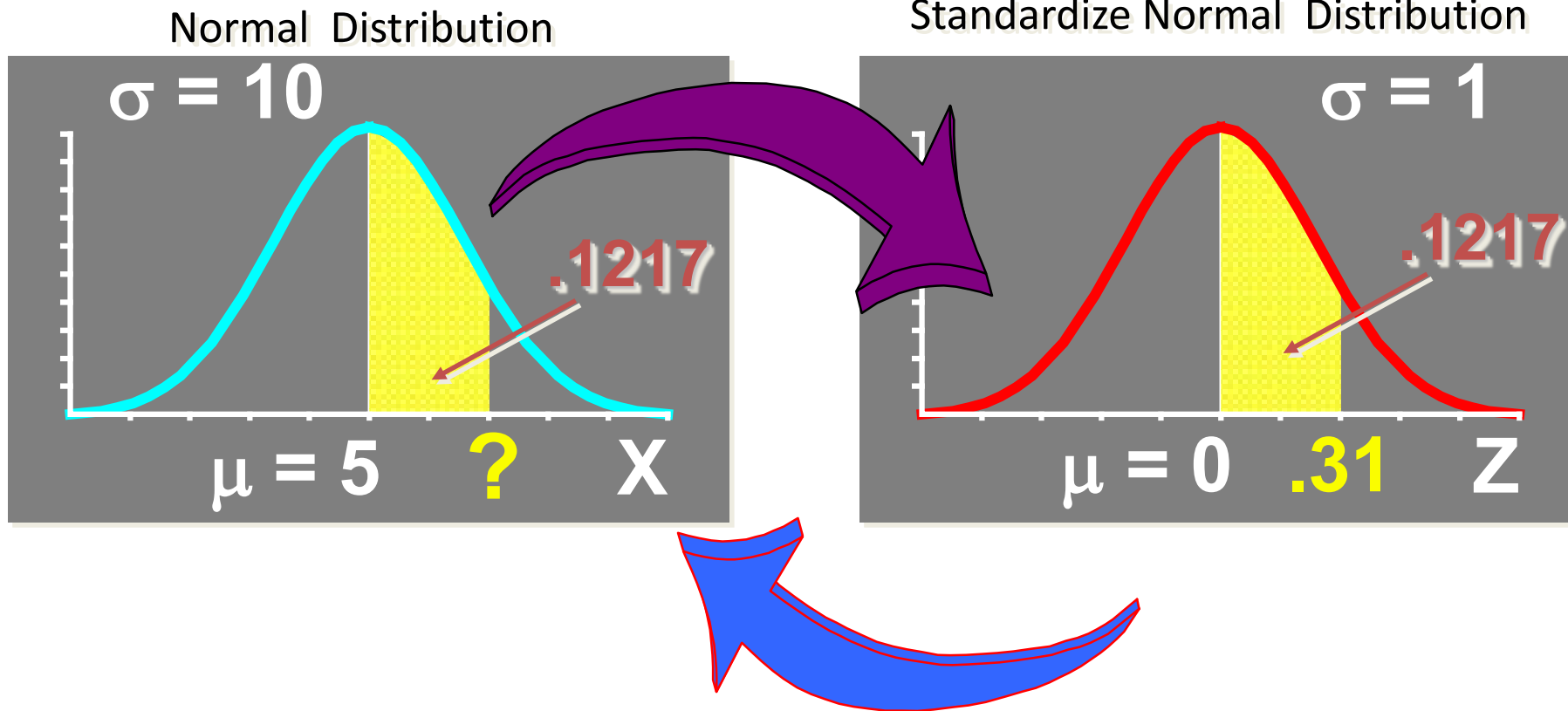
z	.00	.01	.02	.03	.04
0.0	.5000	.5040	.5080	.5120	.5160
0.1	.5398	.5438	.5478	.5517	.5557
0.2	.5793	.5832	.5871	.5910	.5948
0.3	.6179	.6217	.6255	.6293	.6331
0.4	.6554	.6591	.6628	.6664	.6700

z	.00	.01	.02	.03
0.0	.0000	.0040	.0080	.0120
0.1	.0398	.0438	.0478	.0517
0.2	.0793	.0832	.0871	.0910
0.3	.1179	.1217	.1255	.1293
0.4	.1554	.1591	.1628	.1664

Note:

If Z less than mean, then Z value has negative sign.

Finding X Values for Known Probabilities



$$X = \mu + Z \cdot \sigma = 5 + (.31)(10) = 8.1$$

Exercise #7

1. X is a normally distributed variable with mean $\mu = 30$ and standard deviation $\sigma = 4$. Find
 - a) $P(X \leq 40)$
 - b) $P(X \geq 21)$
 - c) $P(18 \leq X \leq 35)$

2. A radar unit is used to measure speeds of cars on a motorway. The speeds are normally distributed with a mean of 90 km/hour and a standard deviation of 10 km/hr. What is the probability that a car picked at random is travelling at more than 100 km/hour?

Exercise #8

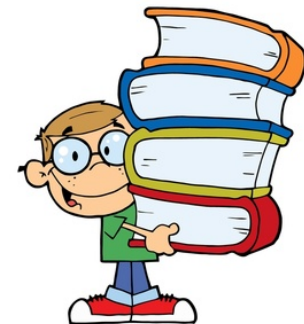
A lecturer commutes daily from his suburban home to his midtown university. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume that the distribution of trip times to be normally distributed.

- a) What is the probability that a trip will take at least $\frac{1}{2}$ hours?
- b) If the office open at 9.00 A.M and the lecturer leaves his house at 8.45 A.M daily, what percentage of the time is he late for work?
- c) If he leaves the house at 8.35 A.M and coffee is served at the office from 8.50 A.M until 9.00 A.M. what is the probability that he missed coffee?
- d) Due to unforeseen incident, the journey takes longer than usual. If 15% is considered as unusual journey time, what is the length of unusual journey time.
- e) Using Binomial distribution, find the probability that 2 of the next 3 trips will take at least $\frac{1}{2}$ hour.

Exercise #9

First year students in UTM are entitled to apply for scholarships. The top 5% of applicants (as measured by CGPA scores) will receive scholarships.

If $\text{CGPA} \sim N(500, 100^2)$, how high does your CGPA score have to be to qualify for a scholarship?

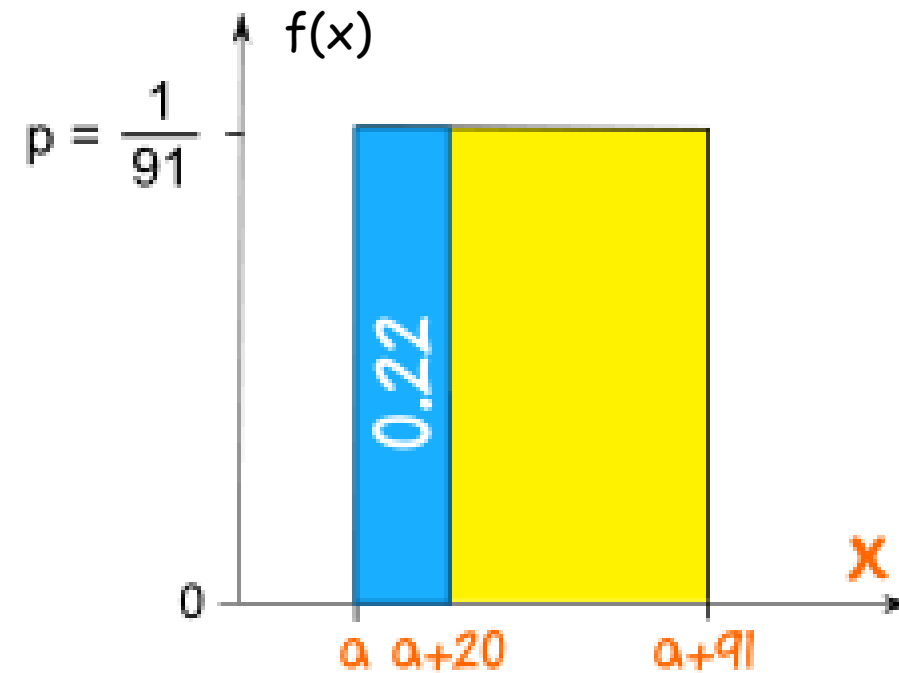


Exercise #10

Base on a survey in 2013, family income is normally distributed with $\mu = \text{RM } 25,000$ and $\sigma = 10,000$. If the poverty level is RM10,000, what percentage of the population lives in poverty?

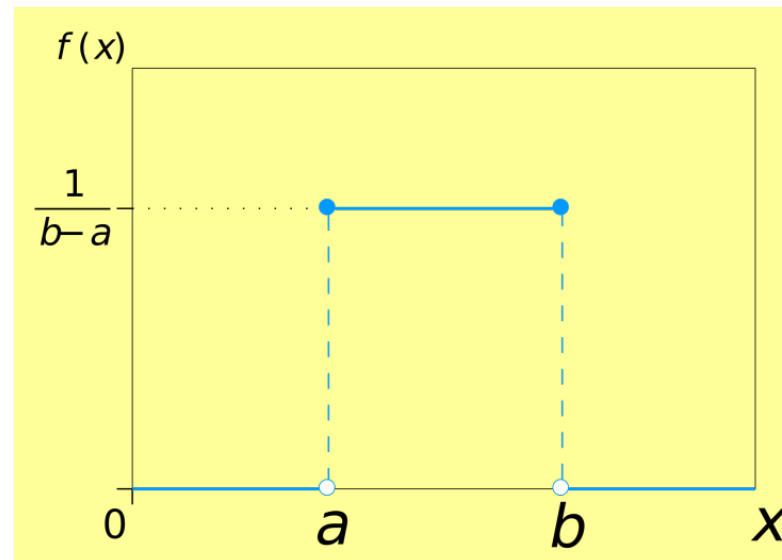


Uniform Distribution



Properties of Uniform Distributions

- Rectangular shape defined by random variable X restricted to a finite interval $[a, b]$ with $f(x)$ has constant density over the interval



Properties of Uniform Distributions

- The uniform random variable X whose density function $f(x)$ is defined by:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Example

A volcano erupts every 91 minutes. You arrive there at random and wait for 20 minutes . What is the probability you will see it erupt?

This is actually easy to calculate, 20 minutes out of 91 minutes is:

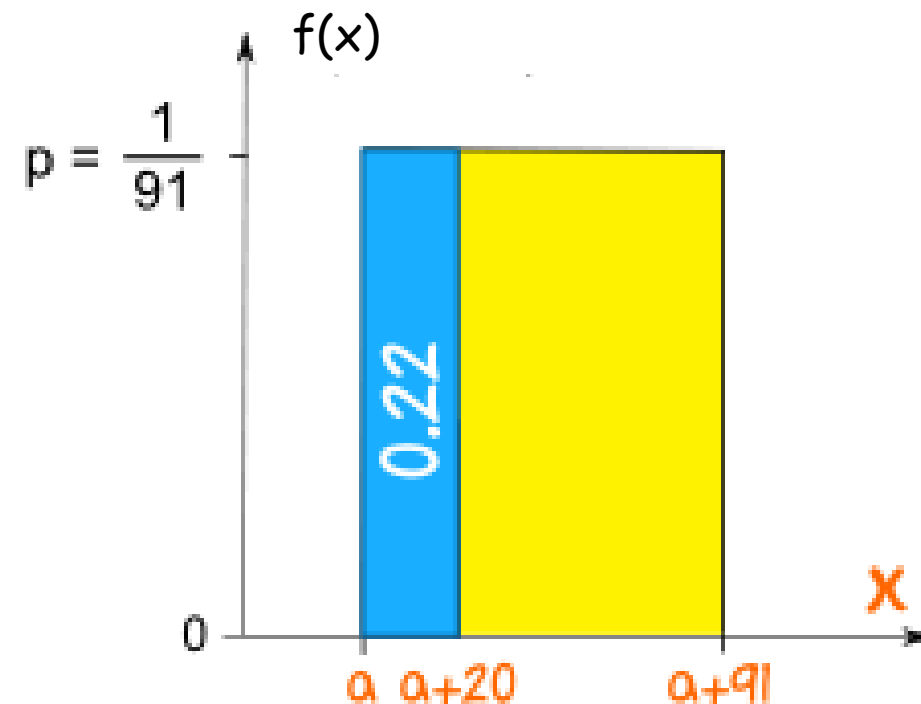
$$p = 20/91 = 0.22 \text{ (to 2 decimals)}$$

But let's use the Uniform Distribution for practice.

Example

- To find the probability between a and $a+20$, find the blue area:

$$\begin{aligned}\text{Area} &= (1/91) \times (a+20 - a) \\ &= (1/91) \times 20 = 20/91 \\ &= 0.22 \text{ (to 2 decimals)}\end{aligned}$$



Example

- So there is a **0.22** probability you will see the volcano erupts.
- If you waited the full **91 minutes** you would be sure (**$p=1$**) to have seen it erupt.
- But remember this is a random thing! It might erupt the moment you arrive, or any time in the 91 minutes.

Exercise #11

- Let x be the amount of time (in minutes) that a particular commuter must wait for a train. Suppose the density function is

$$f(x) = \begin{cases} 0.05 & 0 \leq x \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

- Draw the density curve
- What is the probability that x is between 7 and 12 minutes
- Find the value c for which $p(x < c) = 0.9$